CHEMISTRY

(SCIENCE PAPER – 2)

Maximum Marks: 80

Time allowed: Two hours

1. *Answers to this Paper must be written on the paper provided separately.*
2. *You will **not** be allowed to write during first **15** minutes.*
3. *This time is to be spent in reading the question paper.*
4. ***The time given at the head of this Paper is the time allowed for writing the answers.***

5. ***Section A is compulsory**. Attempt **any four** questions from **Section B**.*
6. *The intended marks for questions or parts of questions are given in brackets [ ].*

Instruction for the Supervising Examiner

Kindly read aloud the Instructions given above to all the candidates present in the Examination Hall.

This paper consists of 16 printed pages.

SECTION A (40 Marks)

*(Attempt **all** questions from this **Section**.)*

Question 1

Choose the correct answers to the questions from the given options. [15]

(Do not copy the question, write the correct answers only.)

(i) Acid + Metal oxide on reaction gives:

(a) base + water

(b) salt + water

(c) base + salt

(d) metal + salt

(ii) A polar covalent compound with two lone pairs of electrons is:

(a) ammonia

(b) water

(c) chlorine

(d) oxygen

(iii) What do you observe when ammonium hydroxide solution is added first dropwise and then in excess to a solution of copper sulphate?

(a) A pale blue precipitate that is insoluble in excess of ammonium hydroxide is seen.

(b) A curdy white precipitate that is soluble in excess of ammonium hydroxide is seen.

(c) A pale blue precipitate is formed that dissolves in excess of ammonium hydroxide to form a deep blue solution.

(d) A curdy white precipitate that is insoluble in excess of ammonium hydroxide is seen.

(iv) Identify the statement(s) that is/are **NOT** true for Group 1 elements:

1. They are good conductors of heat.
2. They are good conductors of electricity.
3. They are strong oxidising agents.
4. They form covalent compounds with non-metals.

(a) Only 1

(b) Only 3

(c) Both 3 and 4

(d) Both 2 and 3

(v) The organic compound which undergoes substitution reaction is:

(a) C_2H_6

(b) C_2H_4

(c) C_2H_2

(d) $C_{10}H_{18}$

(vi) What do you observe when concentrated hydrochloric acid is added to manganese dioxide and then heated?

(a) A colourless gas is liberated that puts out a burning splinter with a 'pop' sound.

(b) A reddish-brown gas is liberated that turns moist potassium iodide paper brown.

(c) A colourless gas is liberated that turns lime water milky.

(d) A greenish yellow gas is liberated that turns moist starch iodide paper blue-black.

(vii) Which of the following occupies the maximum volume at STP?

[At. wt. C=12, O=16]

(a) 3 moles of hydrogen

(b) 1 mole of helium

(c) 64 grams of oxygen

(d) 44 grams of carbon dioxide

(viii) The table below shows the atomic numbers of elements W, X, Y and Z.

Element	Atomic Number
W	4
X	13
Y	7
Z	10

Which of the above elements will combine to form an electrovalent compound?

(a) Y and Z

(b) C and Z

(c) X and Y

(d) X and W

(ix) **Assertion (A):** Noble gases are unreactive under standard conditions.

Reason (R): Noble gases have complete valence shells making them stable and less likely to gain or lose electrons.

(a) (A) is true but (R) is false.

(b) (A) is false but (R) is true.

(c) Both (A) and (R) are true and (R) is the correct explanation of (A).

(d) Both (A) and (R) are true but (R) is not the correct explanation of (A).

(x) The process of dressing (purification) of the ore which involves separation of ore from gangue due to preferential wetting is :

(a) Magnetic Separation

(b) Hydrolytic method

(c) Froth flotation method

(d) Chemical method

(xi) If an element **A** belongs to Period 3 and Group 2, then it will have:

(a) 3 shells and 2 valence electrons

(b) 2 shells and 3 valence electrons

(c) 3 shells and 3 valence electrons

(d) 2 shells and 2 valence electrons

(xii) The vapour density of nitrogen dioxide [At. wt. N = 14, O = 16] is:

(a) 15

(b) 30

(c) 46

(d) 23

(xiii) Metallic and non-metallic elements can both be extracted by electrolysis.
Which of the following will undergo reduction at the electrode?

(a) Bromine

(b) Chlorine

(c) Hydrogen

(d) Oxygen

(xiv) Sodium is in Group I of the Periodic Table and chlorine is in Group VII.

Which statement correctly describes what happens when sodium bonds ionically with chlorine?

(a) Sodium atom gains an electron to form Na^- and chlorine atom loses an electron to form Cl^+

(b) Sodium atom gains an electron to form Na^+ and chlorine atom loses an electron to form Cl^-

(c) Sodium atom loses an electron to form Na^- and chlorine atom gains an electron to form Cl^+

(d) Sodium atom loses an electron to form Na^+ and chlorine atom gains an electron to form Cl^-

(xv) Hydrogen chloride is soluble in water because it is a / an:

(a) base

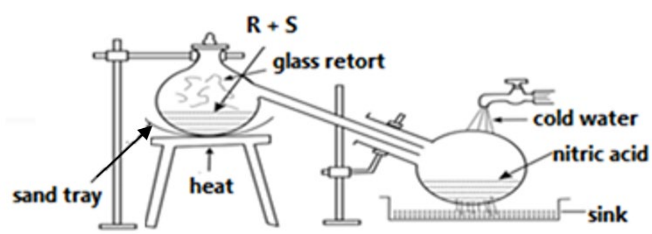
(b) ionic compound

(c) polar covalent compound

(d) non-polar covalent compound

Question 2

(i) The figure given below illustrates the apparatus used in the laboratory preparation of nitric acid. [5]



(a) Name:

1. **R** (a liquid) and

2. **S** (a solid)

(b) Why should we use only glass apparatus in the laboratory preparation of nitric acid?

(c) Write an equation to show how nitric acid undergoes decomposition.

(d) What do you observe when copper is oxidized by concentrated nitric acid?

(ii) Match **Column 1** with **Column 2** without repeating the options from **Column 2**. [5]

	Column 1		**Column 2**
(a)	Ammonia	1.	triple covalent bond
(b)	Carbon tetrachloride	2.	double covalent bond
(c)	Sodium chloride	3.	three single covalent bonds
(d)	Nitrogen	4.	electrovalent bond
(e)	Oxygen	5.	non-polar covalent bond

(iii) Name the following: [5]

(a) A salt which is soluble in hot water, but insoluble in cold water.

(b) An alkali in which both copper hydroxide and zinc hydroxide can be dissolved.

(c) The cation that forms a reddish brown precipitate when its salt solution reacts with a strong alkali.

(d) The law which states that 'equal volumes of all gases, at the same temperature and pressure, have the same number of molecules'.

(e) Two metal wires/plates through which electricity enters and leaves the electrolytic cell during electrolysis.

(iv) Complete the following sentences by choosing the correct word(s) from the brackets. [5]

(a) The IUPAC name for formaldehyde is ___________. *(methanal / ethanal)*

(b) A black solid, ___________ *(manganese dioxide / copper oxide)* reacts with dilute sulphuric acid to form a blue-coloured solution.

(c) The drying agent not used for drying ammonia is__________. *(CaO / P_2O_5)*

(d) The salt formed when iron reacts with dilute HCl is ___________. *($FeCl_3$ / $FeCl_2$)*

(e) Ammonia reacts with excess chlorine to form ___________. *(NCl_3 / NH_4Cl)*

(v) (a) Draw the structural diagram for the following organic compounds. [3]

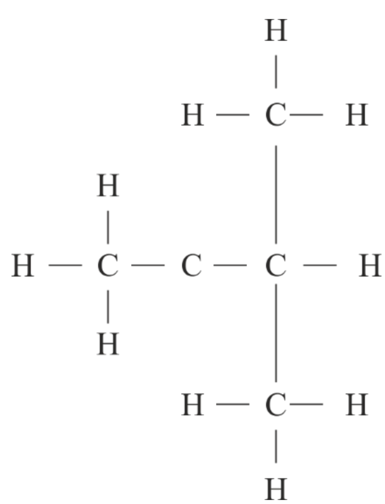
1. ethanal

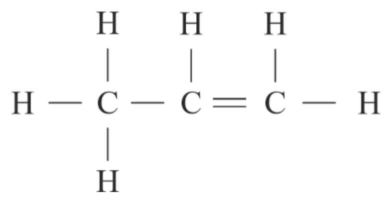
2. acetic acid

3. 2-bromo propane

(b) Give IUPAC name for the following organic compounds. [2]

1.



2. 

SECTION B (40 Marks)

*(Attempt **any four** questions from this **Section**.)*

Question 3

(i) Elements A, B, C & D have the following electronic configurations: [2]

[A = 2, 8, 7 B = 1 C = 2, 5 D = 2, 8, 1]

Answer the following questions:

(a) Identify the element that will form an alkali when reacted with water.

(b) State the formula of the compound formed when D reacts with C.

(ii) The reactions of four different oxides **W, X, Y** and **Z** are given below: [2]

- **W** reacts with hydrochloric acid, but not with sodium hydroxide.
- **X** reacts with both hydrochloric acid and sodium hydroxide.
- **Y** does not react with either hydrochloric acid or sodium hydroxide.
- **Z** reacts with sodium hydroxide, but not with hydrochloric acid.

From the information provided above, identify the oxide which is:

(a) amphoteric

(b) acidic

(iii) 20.2g of potassium nitrate is decomposed by heating according to the equation given below: [3]

$$2KNO_3 \rightarrow 2KNO_2 + O_2$$

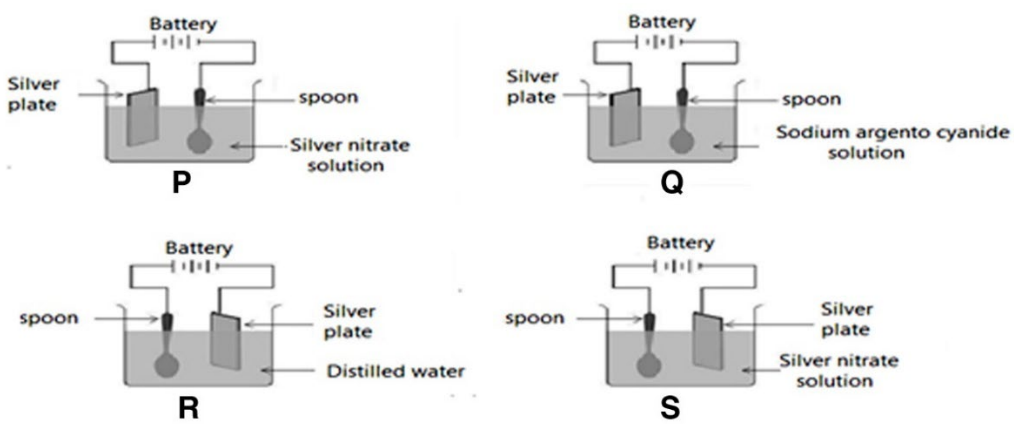
Calculate the following :

(a) Volume of oxygen gas obtained at STP.

(b) Mass of KNO_2 formed.

[At. wt. of K = 39, N=14, O=16]

(iv) A student wanted to electroplate a spoon with silver. He tried four different set-ups to electroplate the spoon as depicted in the diagrams given below. [3]



Answer the following questions based on the above diagrams:

(a) Identify the set-up made by the student which would result in the uniform and smooth electroplating of the spoon with silver.

(b) In which set up will the coating on the spoon be uneven?

(c) Write the reaction taking place at the anode for the correct set-up.

Question 4

(i) Select the compound / element from the box given below for each of the following statements: [2]

H_2	NO_2	NO	O_2

(a) The colourless gas obtained when a metallic nitrate undergoes thermal decomposition.

(b) The gas which on oxidation produces a coloured gas.

(ii) Distinguish between the following: [2]

(a) Hydronium ion and ammonium ion based on their lone pair of electrons.

(b) Solutions of calcium chloride and calcium nitrate using aqueous silver nitrate.

(iii) You are given the following molecular formulae of some hydrocarbons: [3]

C_5H_8	C_5H_{10}	C_8H_{14}	C_7H_{14}	C_6H_6	C_6H_{12}

(a) Which *two* compounds represent unsaturated hydrocarbons having triple bonds?

(b) Identify the formula which represents a cyclic compound.

(iv) Choose the answer from the list of compounds given below: [3]

sodium bisulphite,	ferric oxide,	iron (II) chloride,
sodium sulphate,	aluminium oxide	

(a) A compound which is a major constituent of bauxite.

(b) The compound which will form a dirty green precipitate with NaOH.

(c) An acid salt.

Question 5

(i) The compound 'A' has the following percentage composition by mass, carbon 26.7%, oxygen 71.1%, hydrogen 2.2%. Determine the empirical formula of 'A'. If the relative molecular mass of 'A' is 90, what is the molecular formula of 'A' ? [3]

[At. wt. C=12, H=1, O=16]

(ii) What would you say about the pH of a solution- [3]

(a) in which the concentration of H^+ ions is equal to concentration of OH^- ions?

(b) which evolves hydrogen sulphide when added to ferrous sulphide?

(c) which produces a pungent smelling gas when heated with an ammonium salt?

(iii) Answer the following questions based on the information provided for the elements **P, Q** and **R**. [4]

- **P** : atomic number = 19
- **Q** : electronic configuration = 2, 6
- **R** : has a tendency to gain one electron

(a) Write the electronic configuration of element **P**.

(b) What is the valency of element **Q** ?

(c) Is element **R** a metal or a non- metal ?

(d) Which period does **P** belong to ?

Question 6

(i) State **true** or **false** for the following statements : [3]

(a) Brass is an alloy which contains a non-metal as one of its constituents.

(b) Duralumin contains copper.

(c) Vapour density of gas is twice its molecular mass.

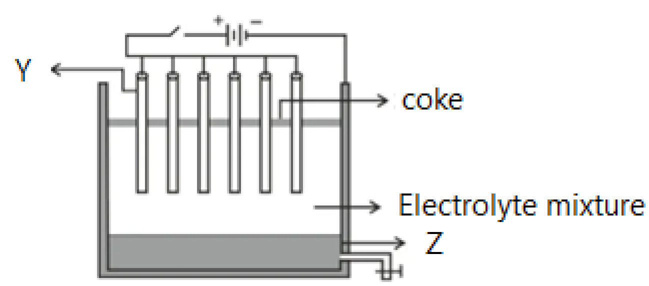
(ii) Deepak spilled some concentrated nitric acid accidentally in the laboratory. To neutralize it, he sprinkled sodium carbonate powder on it. [3]

(a) Write the equation for the reaction that took place when he sprinkled sodium carbonate on nitric acid.

(b) Name the gas evolved in the reaction above.

(c) Write the equation for the reaction that would take place if Deepak had sprinkled carbon over concentrated nitric acid.

(iii) The following sketch illustrates the process of conversion of Alumina to Aluminum. Study the diagram and answer the following: [4]



(a) Name the constituent of the electrolyte mixture which has a divalent metal in it.

(b) Write the reactions taking place at the electrodes:

1. **Y** (anode) and

2. **Z** (cathode) respectively.

(c) Give *one* use of sprinkling coke on the electrolyte.

Question 7

(i) Which property of concentrated H_2SO_4 is used in each of the following cases? [2]

(a) For the production of HCl gas when it reacts with a metal chloride.

(b) For the conversion of blue copper sulphate crystals to white anhydrous copper sulphate.

(ii) Identify the **anion** present in the following compounds: [2]

(a) Compound 'C' on heating with dilute hydrochloric acid liberates a gas which turns lime water milky but has no effect on potassium dichromate solution.

(b) Compound 'D' on heating with concentrated sulphuric acid liberates a gas which when passed through silver nitrate solution gave a white precipitate, insoluble in dilute nitric acid.

(iii) From the list given, choose the method which is most suitable for the preparation of the following salts? [3]

- E - Neutralization by titration
- F - Precipitation
- G - Simple displacement
- H - Direct combination

(a) Zinc sulphate

(b) Silver chloride

(c) Ammonium nitrate

(iv) Metal **X** is low in the reactivity series and it is obtained by the electrolysis of its bromide. [3]

Complete the following blanks by choosing the correct word from the options given in the brackets:

(a) Metal **X** is __________. *(lead / sodium)*

(b) The bromide is in the __________ state. *(solution / molten)*

(c) The sulphate of metal **X** will have the formula _________. *(X_2SO_4 / XSO_4)*

Question 8

(i) Arrange the following as per the instructions given in the bracket. [3]

(a) Na, Al, P, Si, Mg (increasing order of metallic character)

(b) Cl, F, Br, I (increasing order of electronegativity)

(c) He, Ar, Ne, Ca (increasing order of the number of shells)

(ii) Study the part of the Periodic Table given below and answer the questions that follow: [3]

(Do not identify the elements)

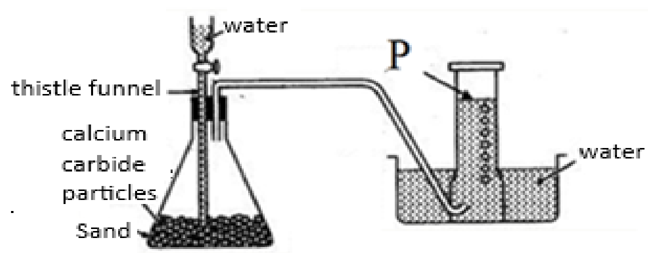
1	**2**	**13**	**14**	**15**	**16**	**17**	**18**
					J		Q
	L						
M			O			P	
N							G

(a) Draw the dot (.) and cross (×) structure of the compound formed between elements **L** and **P**.

(b) Which element has the highest ionization potential?

(c) Element **N** is more reactive than **M**. Justify.

(iii) The following diagram shows the laboratory preparation of a gaseous hydrocarbon **(P)** containing *two* carbon atoms. [4]



(a) Identify the gaseous product **P**.

(b) Give the general formula of the homologous series which **P** belongs to.

(c) Write a balanced equation for the reaction that takes place.

(d) State the type of reaction that **P** undergoes when it reacts with hydrogen.